Shresth Rohan

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Summary

CS (Data Science) undergraduate building machine learning, full-stack and embedded systems: a from-scratch DSL compiler, a real-time computer vision attention monitor and a patent-filed IoT system.

Education

Vellore Institute of Technology — Vellore, Tamil Nadu
B.Tech. in Computer Science, Specialization in Data Science; CGPA: 8.76/10 — *Jul 2024 – May 2028*

St. Michael's High School — Patna, Bihar
Class XII (CBSE) – 90.2% — *May 2024*

Don Bosco Academy — Patna, Bihar
Class X (ICSE) – 96.8% — *May 2022*

Projects

CleanScript – Data Cleaning DSL Compiler | *Python, FastAPI, Pandas, React* — Oct 2026 – Present

- Designed a 10-command domain-specific language for cleaning CSV datasets and built its compiler from scratch
- Hand-wrote the **lexer**, **recursive-descent parser**, schema-aware **semantic analyzer**, **IR** and **code generator**
- Executed programs on a sandboxed IR interpreter (no eval/exec), checked by **194 pytest unit tests**
- Benchmarked **1M rows cleaned in 2.7 s**, output identical to hand-written Pandas, plain Python and SQLite
- Built a full-stack React + FastAPI app that visualizes each compiler stage and auto-suggests dataset fixes

FocusGuard | *Python, PyTorch, ONNX, Flask, React, Node.js, MongoDB* — Feb 2026 – Present

- Built a real-time **computer vision** focus monitor using **MobileNetV2 transfer learning** on webcam input
- Served a custom **ONNX** deep learning model via a **Flask REST API** with MediaPipe face detection
- Developed a React analytics and productivity dashboard backed by a **Node.js/Express API** and **MongoDB**
- Implemented a **Chrome extension** using MutationObserver to hide distracting feeds on YouTube and Instagram

Smart Adaptive Street Lighting (Patent Filed) | *ESP32, C++, IoT Sensors* — Apr 2025 – Present

- Built an **ESP32 embedded system** fusing LDR, rain, sound and ultrasonic sensors for adaptive lighting
- Implemented **dynamic brightness control** with siren detection to save energy and prioritize emergency vehicles
- Designed a **servo-assisted ultrasonic scanner** for vehicle detection and real-time overspeed alerts
- **Filed an Indian patent application** for the integrated multi-sensor smart-city lighting design

Technical Skills

Languages: Python, Java, C/C++, SQL, JavaScript, HTML/CSS
Frameworks: React, Node.js, Express, FastAPI, Flask, Spring Boot, Vite
AI/ML & Data: Machine Learning, Deep Learning, Computer Vision, PyTorch, ONNX, MediaPipe, Pandas, NumPy
Databases: MongoDB, PostgreSQL, SQLite
Developer Tools: Git, Docker, Pytest, Vitest, VS Code, Arduino IDE
Concepts: Data Structures & Algorithms, Object-Oriented Programming, Compiler Design, REST APIs, Unit Testing

Achievements & Leadership

- **State Rank 3**, **International Rank 239** – SilverZone International Informatics Olympiad
- **Zonal Rank 1**, **International Rank 85** – SOF International English Olympiad
- **Volunteer, Karo Sambhav Movement** – supported e-waste collection and recycling awareness drives
- **Volunteer Instructor, VIT Outreach Program** – taught computers and coding to middle-school students